1st Place: Aspect 5 – Use mathematical outcomes in a variety of formats in relation to a situation; compare or evaluate two or more representations

- **Analysis**: This is the most central competency required by the question. The problem provides data in a table (Music: 550 MB, Photos: 338 MB, Free space: 112 MB) and four pie chart options (A, B, C, D). Students must translate the numerical information from the table into a judgment about graphical representation, compare and evaluate the different graphs, and select the one that correctly reflects the proportional relationships in the data.
- **Manifestation in the Question**: Students must understand the size relationships among the three data categories in the table (550 > 338 > 112) and determine the correct order of sector sizes in a pie chart. They must then eliminate options that do not match these proportional relationships.

2nd Place: Aspect 1 – Interpret a mathematical result back into the real-world context

- **Analysis**: The context is "transferring content from an old storage device to a new one." The calculation result (total data: 550+338+112 = 1000 MB) and the graph selection must be interpreted within this real-world scenario: the new device (2000 MB) is half full, with the other half free.
- **Manifestation in the Question**: Students need to calculate that the total used space is 1000 MB and the new device's total capacity is 2000 MB. Therefore, the "free space" should be exactly half of the total capacity (1000 MB), not the 112 MB from the old device's data. This requires students to connect the mathematical result (1000 MB used, 1000 MB free) with the graphical representation where the "free space" sector should represent 1/2 of the circle.

3rd Place: Aspect 2 – Identify whether a mathematical result or conclusion makes sense given the context

- **Analysis**: Students must judge whether the selected graph reasonably represents the status "after migration," rather than directly copying the proportions from the old device's data.
- **Manifestation in the Question**: Directly plotting the old proportions (550:338:112) is a trap. The correct reasoning is: after migration, the used data remains 1000 MB, but the total capacity becomes 2000 MB, so free space = 2000 - 1000 = 1000 MB. Thus, the "free space" sector should occupy 50% of the chart. Students must recognize and eliminate options that do not correctly reflect this change in the new device's capacity.

4th Place: Aspect 9 – Understand the relationship between context and solution, and assess feasibility

- **Analysis**: Students must understand how the context ("migration to a higher-capacity device") changes the way data is represented graphically and evaluate the feasibility of each graph option.
- **Manifestation in the Question**: The correct option must satisfy both: 1) the relative size order of the three data categories matches the table (Music > Photos > Free? Note: In the new device, "Free" is actually 1000 MB, larger than both Music and Photos, so the order becomes: Free > Music > Photos); 2) the free space sector occupies 1/2 of the circle. Students must comprehensively assess which option meets these conditions.

5th Place: Aspect 6 – Use knowledge of how the real world impacts outcomes and calculations to make contextual judgments

- **Analysis**: The real-world context (device replacement, doubled capacity) directly influences the interpretation of the data and the choice of graph.
- **Manifestation in the Question**: Students cannot mechanically treat the table data as the final graphical data. They must adjust their understanding based on the real-world operation: "the new device is initially empty, so free space is recalculated after migration." Free space is no longer the original 112 MB, but the new capacity minus the used data.

6th Place: Aspect 7 – Construct and communicate explanations and arguments

- **Analysis**: Although the question only asks to circle an option, a correct selection necessarily involves an internal reasoning process.
- **Manifestation in the Question**: To choose correctly, students must mentally construct an argument, such as: "Total data = 550+338+112 = 1000 MB. New device capacity = 2000 MB, so free space = 1000 MB. Therefore, free space is 50% of the total, and 1000 > 550 > 338, so the free space sector is the largest and occupies a semicircle. Only option X satisfies this."

第1位：方面 5 - 使用多种数学表达形式解释情境，并比较或评估这些表达

- **分析**：这是本题最核心的素养要求。题目提供了表格数据（Music: 550 MB, Photos: 338 MB, Free space: 112 MB）和四个饼图选项（A、B、C、D），要求学生将表格中的数值信息转换为对图形表示的判断，并在不同图形之间进行比较、评估，选出正确反映数据比例关系的那一个。
- **题目体现**：学生必须理解表格中三类数据（音乐、照片、空闲空间）的大小关系（550 > 338 > 112），并判断它们在饼图中应有的扇形面积比例顺序，进而排除不符合比例关系的选项。

第2位：方面 1 - 将数学结果解释回现实情境

- **分析**：题目背景是“将旧存储设备的内容迁移到新设备”，计算结果（总数据量为 550+338+112=1000 MB）和图形选择必须放回这个现实情境中理解，即新设备（2000 MB）中，已用空间占一半，剩余一半为空闲。
- **题目体现**：学生需计算出总占用空间为 1000 MB，新设备总容量为 2000 MB，因此“空闲空间”应恰好为总容量的一半（1000 MB），而不是表格中的 112 MB（旧设备数据）。这要求学生将数学计算（1000 MB 已用，1000 MB 空闲）与图形中“空闲部分占比应为 1/2”对应起来。

第3位：方面 2 - 判断数学结果在情境中是否合理

- **分析**：学生需要判断所选图形是否合理反映“迁移后”的状态，而不是直接照搬旧设备的数据比例。
- **题目体现**：直接按旧设备比例（550:338:112）画图是陷阱。正确思路是：迁移后，已用数据仍是 1000 MB，但总容量变为 2000 MB，所以空闲空间 = 2000 - 1000 = 1000 MB。图形中“空闲空间”占比应为 50%。学生需识别并排除那些未正确反映新设备容量变化的选项。

第4位：方面 9 - 理解问题情境与数学解的关系，评估可行性

- **分析**：学生需理解“迁移到更大容量的新设备”这一情境如何改变了数据的图形表示方式，并评估各选项图形是否可行。
- **题目体现**：正确选项必须同时满足：1）三类数据的相对大小关系与表格一致（音乐 > 照片 > 空闲？这里需注意：新设备中“空闲”实际是 1000 MB，大于照片和音乐，因此顺序变为：空闲 > 音乐 > 照片）；2）空闲部分占比为 1/2。学生需综合评估各选项是否满足这些条件。

第5位：方面 6 - 根据现实影响调整数学过程与判断

- **分析**：现实情境（设备更换、容量翻倍）直接影响对数据的理解和图形的选择。
- **题目体现**：学生不能机械地将表格数据视为最终图形数据，必须根据“新设备完全空，迁移后空闲空间重算”这一现实操作调整理解：空闲空间不再是原来的 112 MB，而是新容量减去已用数据。

第6位：方面 7 - 构建解释与论证

- **分析**：虽然题目只要求圈出选项，但正确的选择背后必然包含一个内在的推理过程。
- **题目体现**：为了选出正确选项，学生需要在思维中构建一个论证，例如：“总数据量 = 550+338+112=1000 MB，新设备容量 2000 MB，所以空闲 = 1000 MB。因此空闲占比 50%，且 1000 > 550 > 338，所以图形中空闲部分最大，且占半圆。只有选项X满足。”